Principle of Communication System		
Lab #:	**06**	
Lab Title:	Amplitude Modulation Suppressed Carrier (SC) Implementation using USRP.	
Submitted by:		
Names		**Registration #**
Sumaira Safeer		FA22-BCE-019

Rubrics name & number		Marks	
		In-Lab	**Post-Lab**
Engineering Knowledge	***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.*		
Problem Analysis	***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.*		
Design	***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.*		
Modern Tools Usage	***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.*		
Individual and Teamwork	***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.*		
	R13: Management of Team Work: *Ability to appreciate, understand and work with multidisciplinary team members.*		

Rubrics #	**R2**	**R6**	**R8**	**R9**	**R12**	**R13**
In Lab						
Post- Lab						

LAB: 06

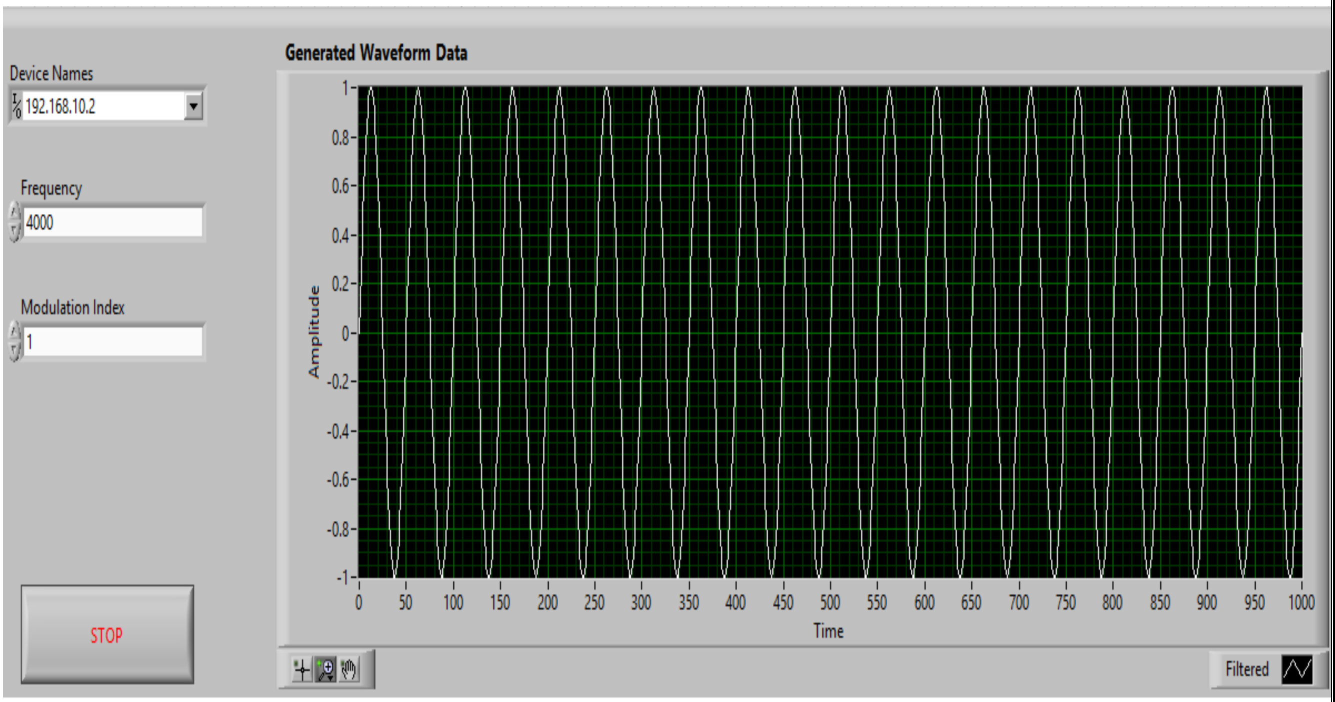
Topic: "Amplitude Modulation Suppressed Carrier (SC) Implementation using USRP."

Objective:

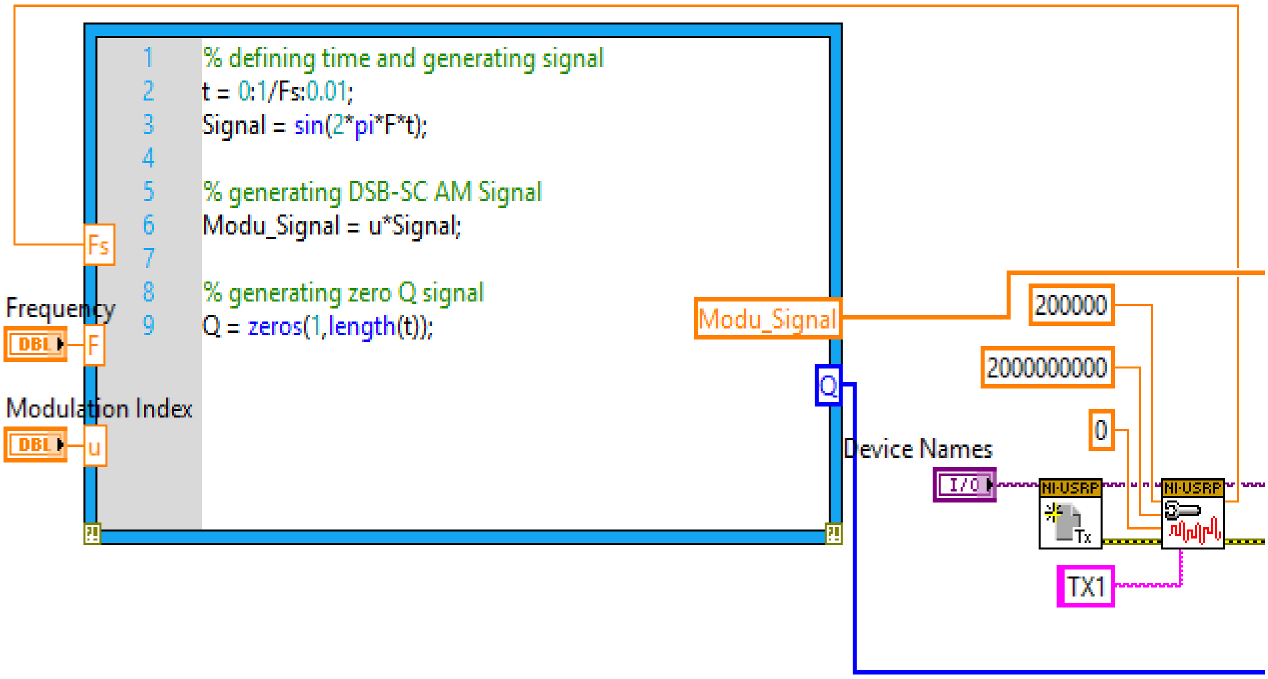
The main objective of this lab is AM (SC) implementation using USRP and LabVIEW.

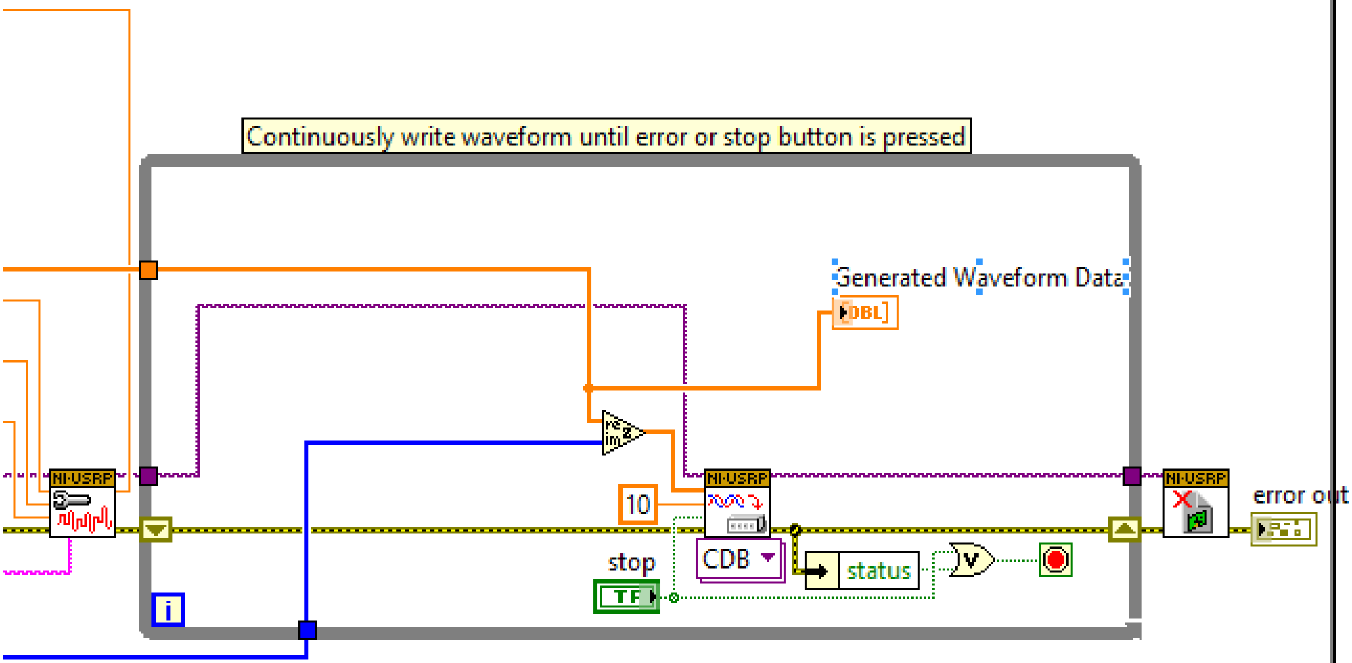
AM Transmitter:

- **FRONT PANEL:**



<u>Block Diagram:</u>



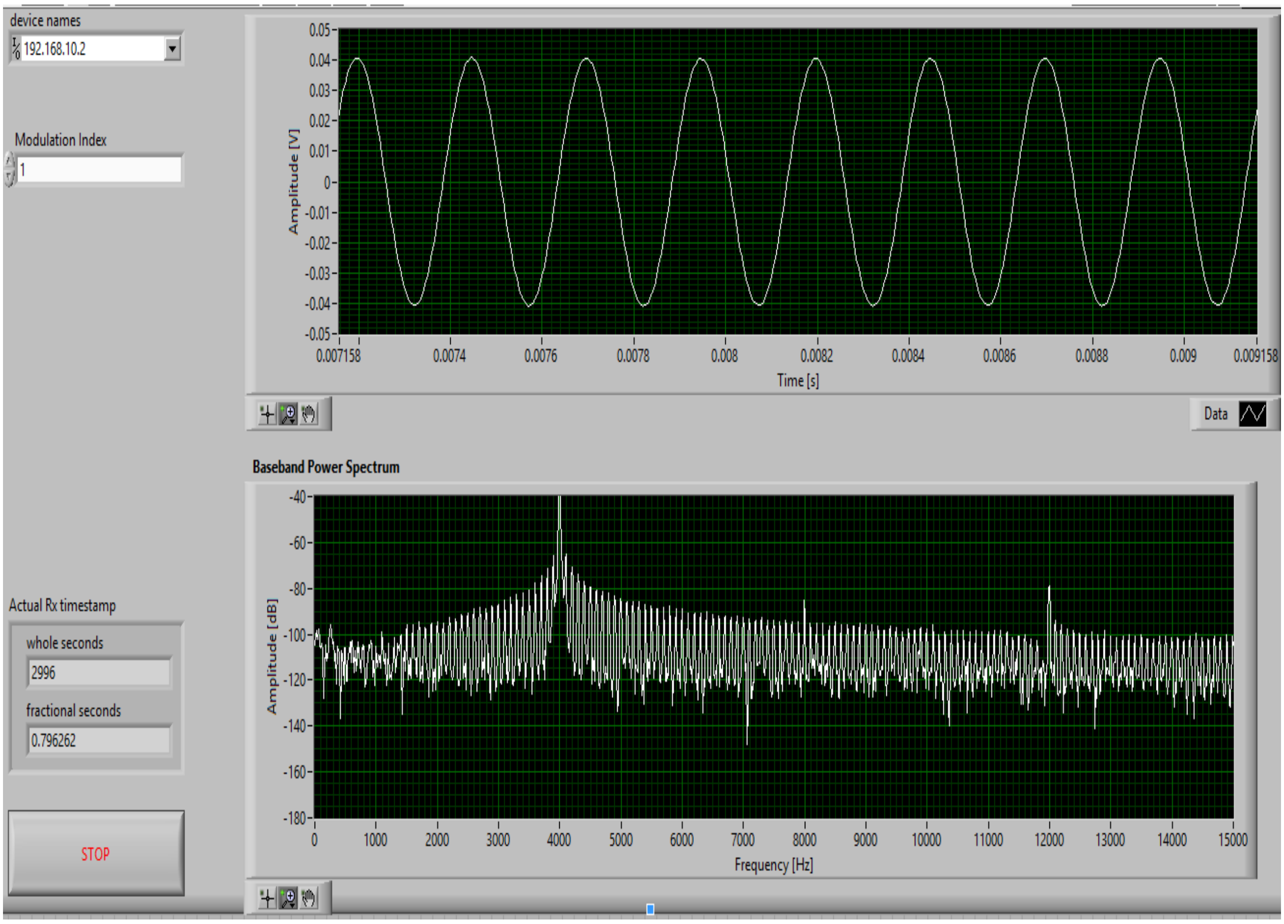


Conclusion for AM (SC) Transmitter:

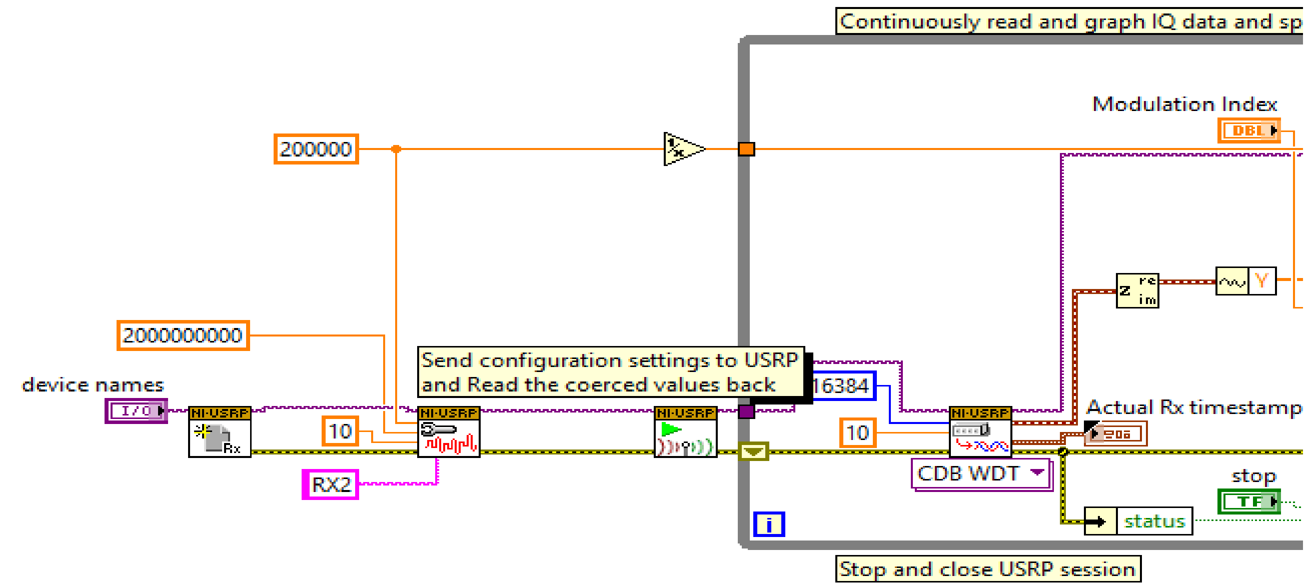
The AM (SC) transmitter successfully generates and transmits a single sideband modulated signal using USRP and LabVIEW. This implementation demonstrates the efficiency of single sideband modulation in conserving bandwidth while maintaining signal integrity, showcasing the capabilities of software-defined radio in modern communication systems.

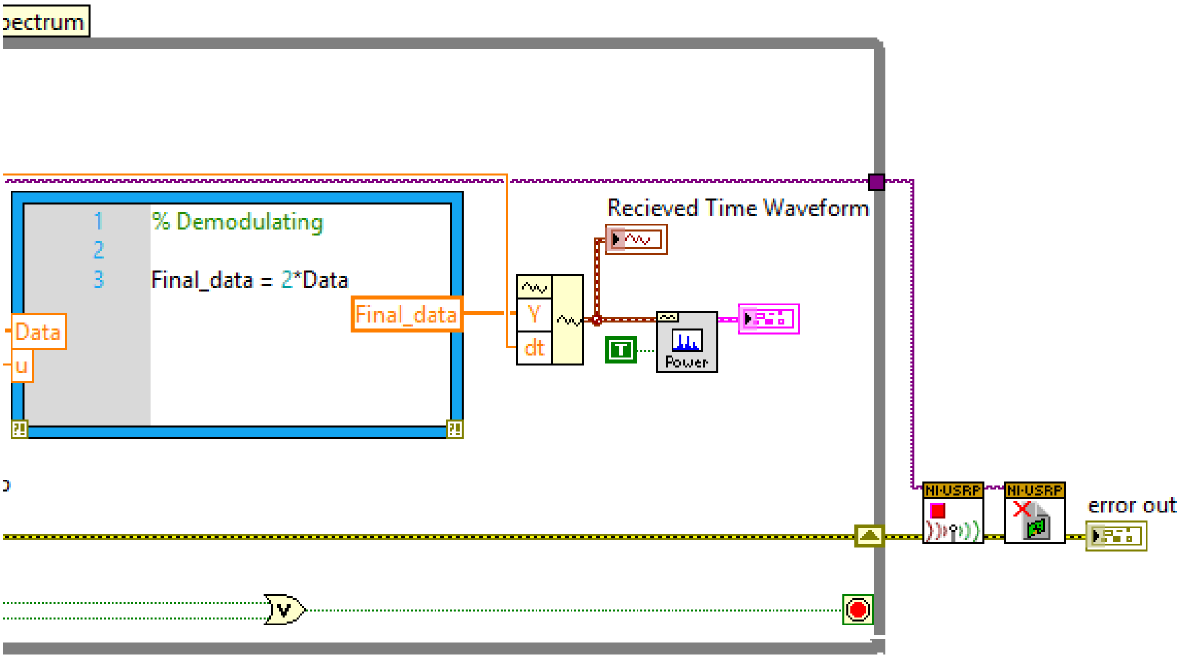
AM Receiver:

- **FRONT PANEL:**



Block Diagram:





Conclusion for AM (SC) Receiver:

The AM (SC) receiver effectively captures and demodulates the transmitted single sideband signal, successfully reconstructing the original audio. This project highlights the importance of advanced signal processing techniques and the reliability of USRP hardware in achieving high-quality wireless communication, paving the way for further exploration in software-defined radio applications.